

Budget — AKG-E Full Proposal (DRAFT)

(DRAFT — for PI / lab-admin review; dollar amounts are placeholders)

Project: Agentic Knowledge Graphs for AI-Accelerated Engineering with Simulation-Grounded Validation

PI: Kereshmeh Afsari **Co-PI:** Chris Brown **Student Researcher:** Jason Cusati

Sponsor: Amazon–Virginia Tech Initiative, CFP 2026–2027

Period: 12 months (3-month aggressive build + 9-month dissemination / scaling)

Category	Amount
Personnel	
GRA #1 — Cusati (12 mo, 50% effort; concentrated during 3-mo build)	\$23,000
Stipend	\$18,000
Tuition	\$4,000
Benefits / fringe	\$1,000
GRA #2 — TBD (12 mo, 50% effort; concentrated during 3-mo build)	\$23,000
Stipend	\$18,000
Tuition	\$4,000
Benefits / fringe	\$1,000
Faculty summer salary — Afsari, 1 mo (optional)	\$14,000
Materials and Computing	
LLM API tokens (development + benchmark runs)	\$15,000
AWS Bedrock + cloud compute (KG hosting, serving, scaling experiments)	\$10,000
Simulation software / FEA licenses (FEniCS open-source + utilities)	\$2,000
Travel	
Conference venue #1 (AI for engineering / agentic AI)	\$3,000
Conference venue #2 (construction / civil engineering)	\$3,000
Amazon collaboration travel (Seattle / Arlington)	\$2,000
Total Direct Costs	\$95,000
Indirect Costs (gift research — no F&A applied)	\$0
Total Project Cost	\$95,000

Justification — LLM API Tokens. AKG-E is, by construction, a high-volume LLM workload: each iteration of the propose → VVUQ → feedback loop spends tokens on retrieval-augmented generation over the curated knowledge graph, on structured graph queries that surface code provisions and physical-law constraints, on simulation feedback ingestion (FEA / digital-twin / PDE solver outputs returned to the agent), and on the refinement step that consumes gate-emitted violations to produce the next candidate artifact. The benchmark evaluation amplifies this footprint substantially: three baselines (B0 Claude alone; B1 Claude + graph-RAG; B2 full AKG-E) are run across three engineering domains (construction, transportation, energy), with $n \geq 5$ random seeds per cell and bootstrap resampling for 95% confidence intervals on the headline correctness, calibration, and efficiency metrics. The full benchmark therefore drives thousands of multi-turn agent runs at production-quality model tokens, and the LLM API line is sized to support both day-to-day development and full benchmark execution without throttling. Separating this line from cloud compute prevents the token spend from being absorbed into a generic AWS line item where Amazon reviewers and VT lab admin cannot audit it independently.

Notes for reviewers: Total ask of \$95,000 sits within the Amazon-VT CFP guidance (“~\$100K range, funded as gift research”). The award is gift research, so no F&A indirect cost is applied; if Amazon elects to rescope as sponsored research, F&A will be added at the VT-applicable rate and the personnel lines adjusted accordingly. Dollar amounts above are draft estimates pending PI sign-off; final GRA stipend and tuition rates for the 2026–2027 academic year are subject to VT lab-admin confirmation. The faculty summer salary line is optional and may be zeroed at the PI’s discretion (reducing total to \$81,000).